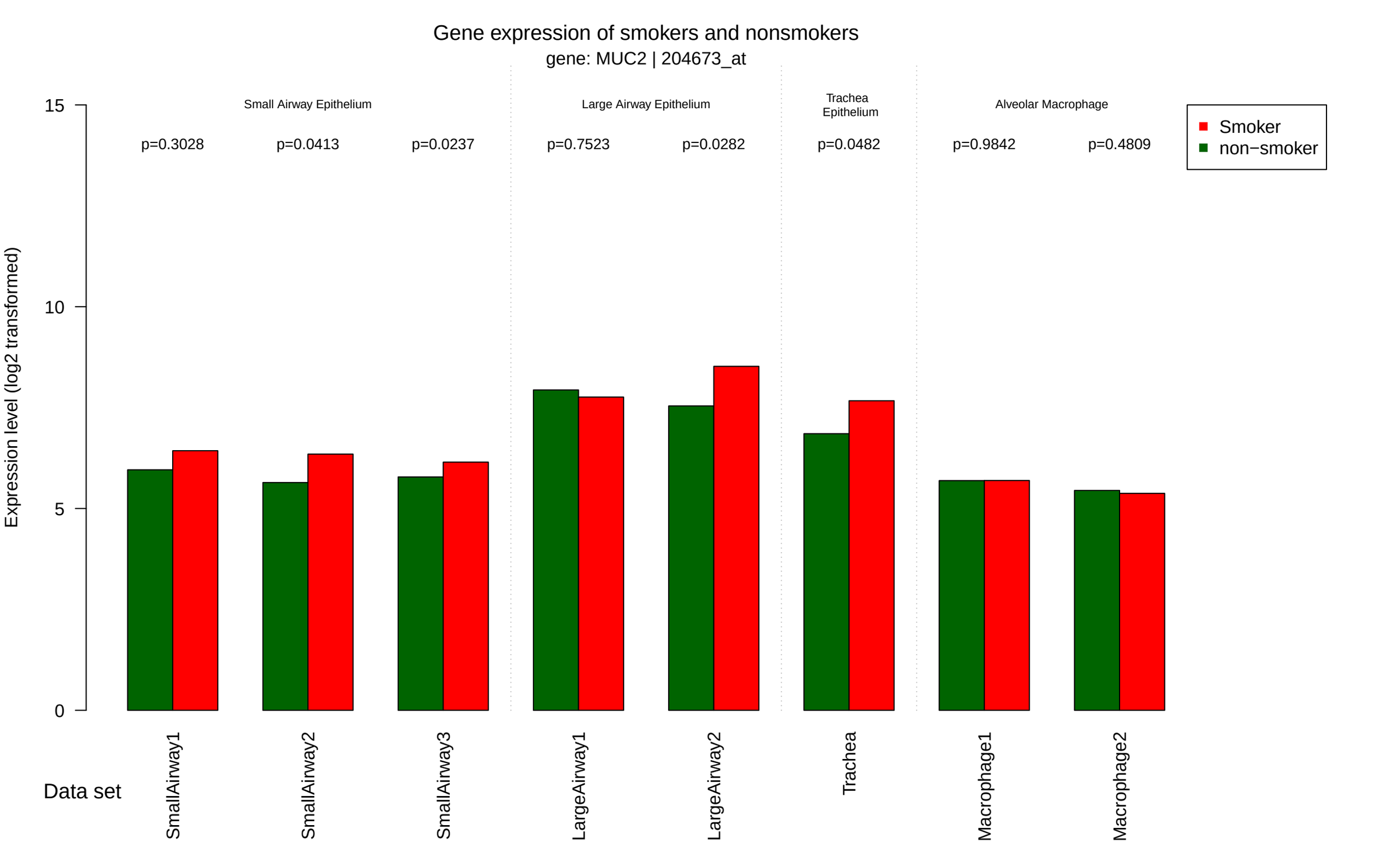
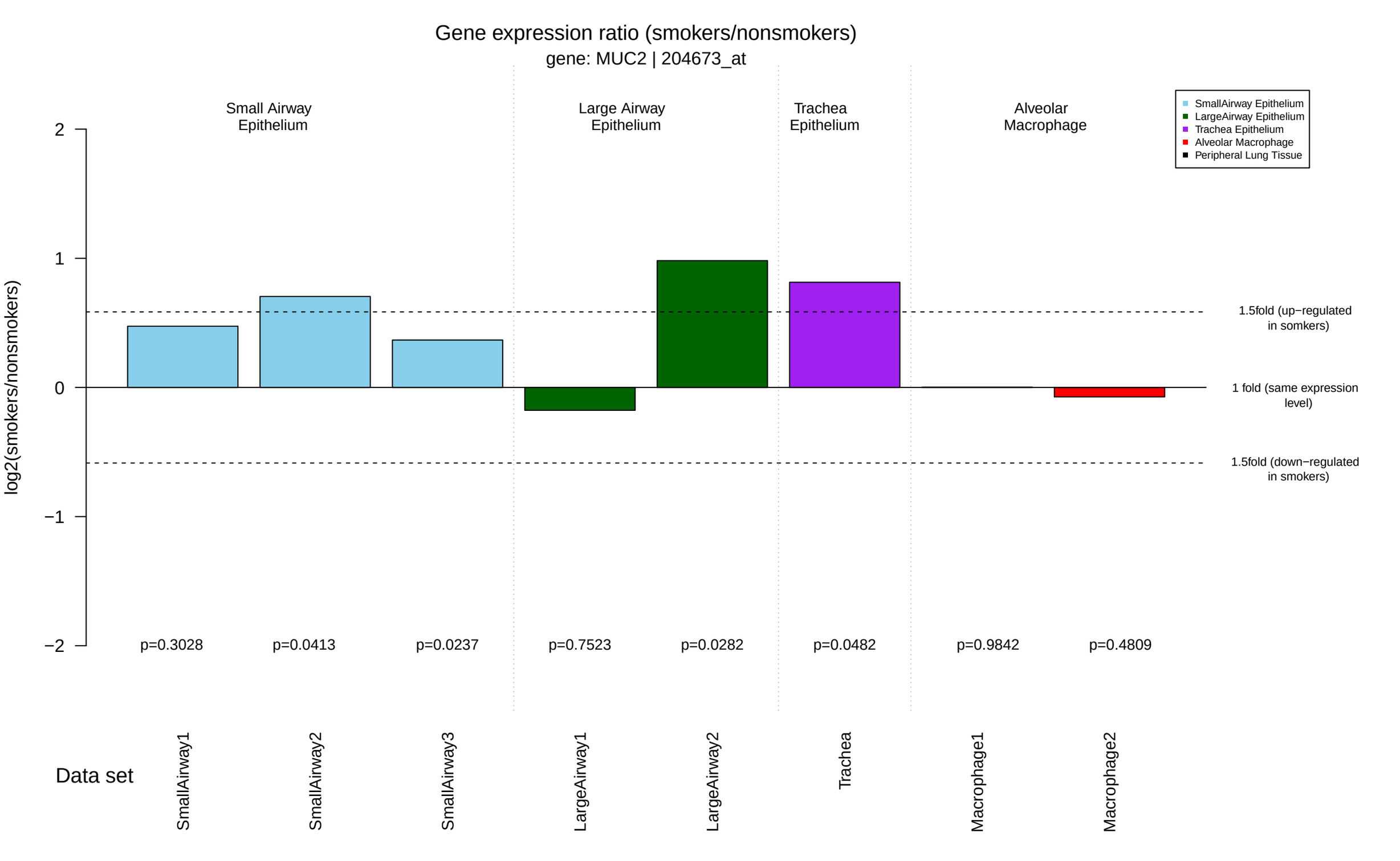
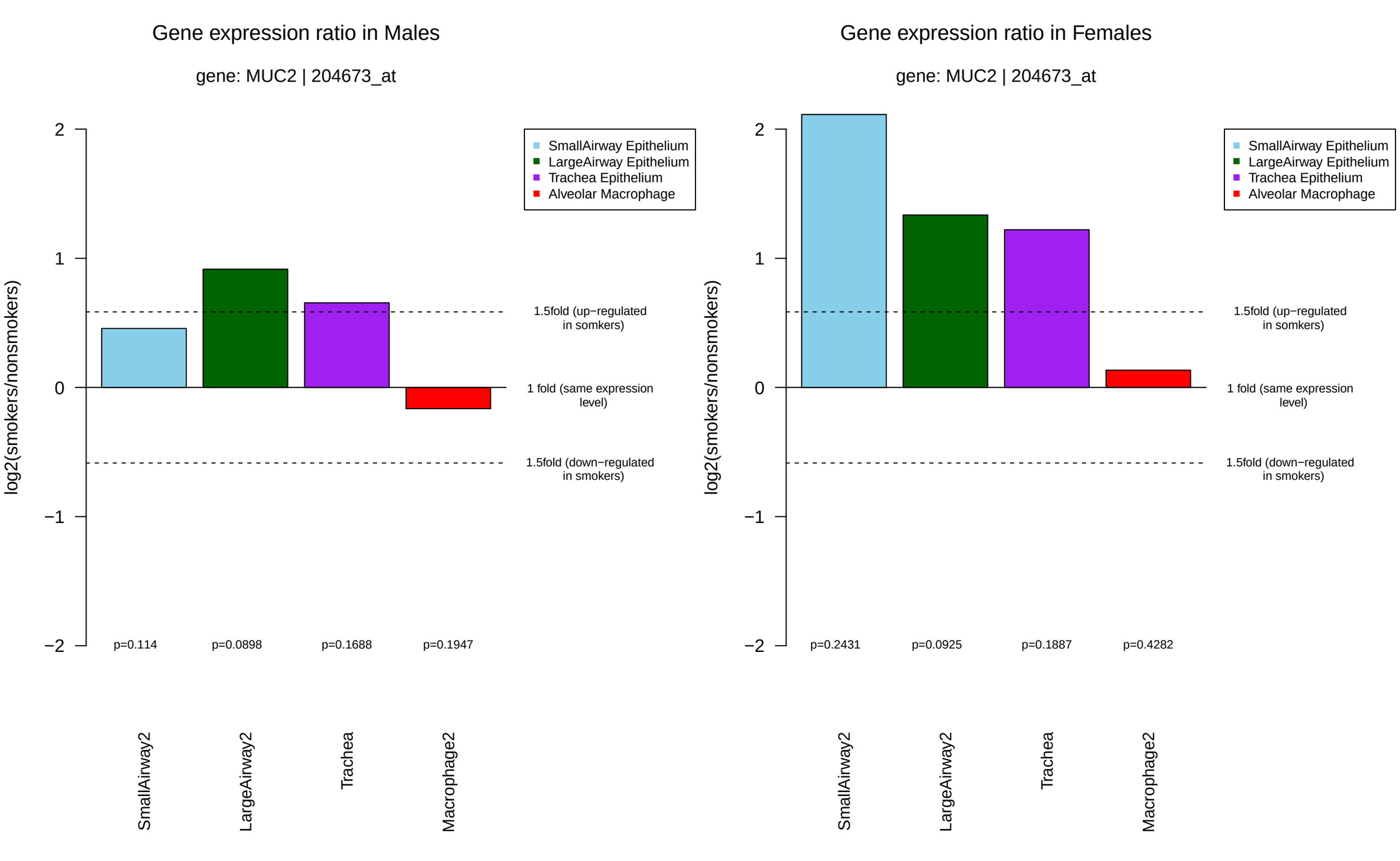
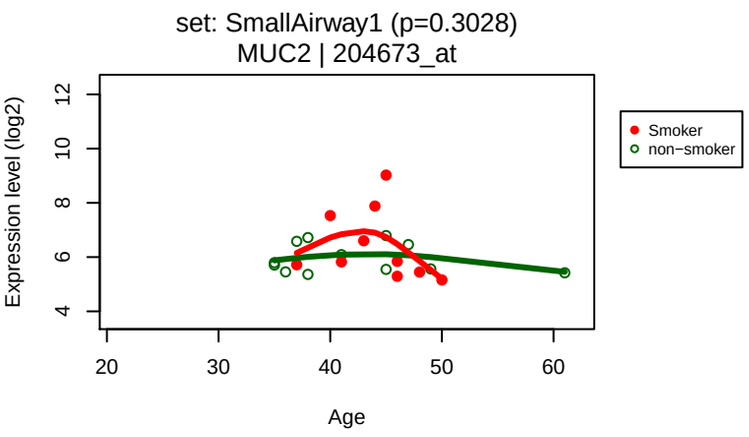
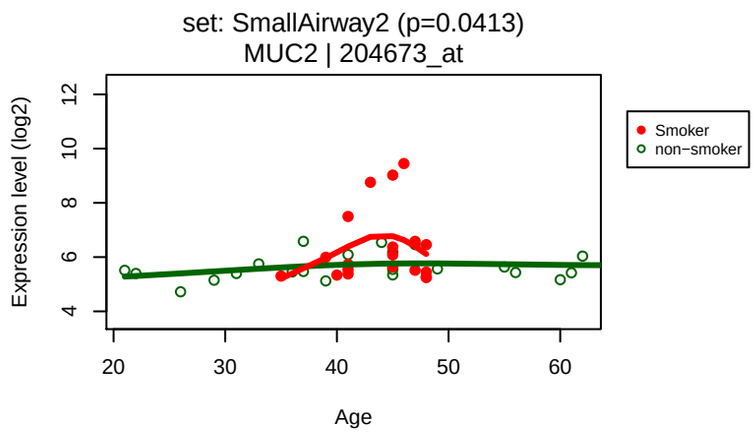
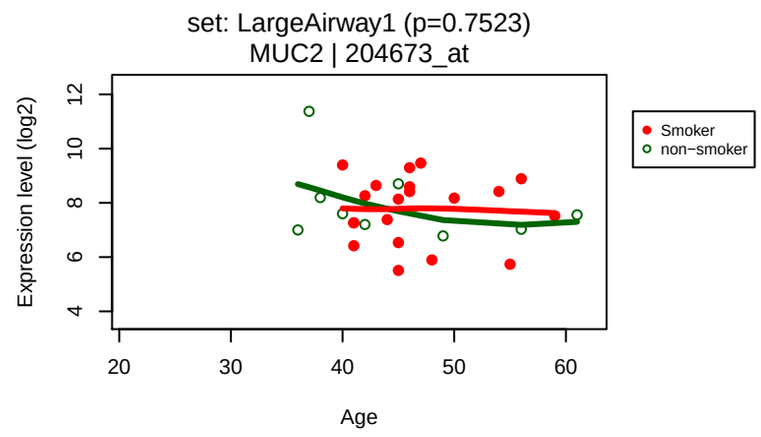

Gene expression of smokers and nonsmokers
gene: MUC2 | 204673_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3028
p=0.0413
p=0.0237
p=0.7523
p=0.0282
p=0.0482
p=0.9842
p=0.4809
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

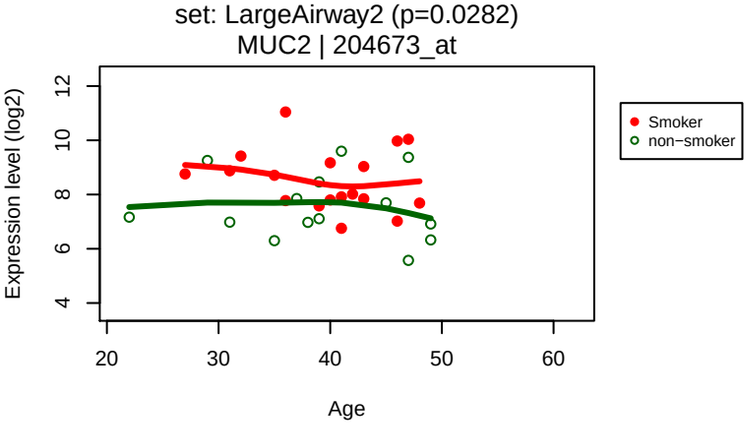

Gene expression ratio (smokers/nonsmokers)
gene: MUC2 | 204673_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3028
p=0.0413
p=0.0237
p=0.7523
p=0.0282
p=0.0482
p=0.9842
p=0.4809
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

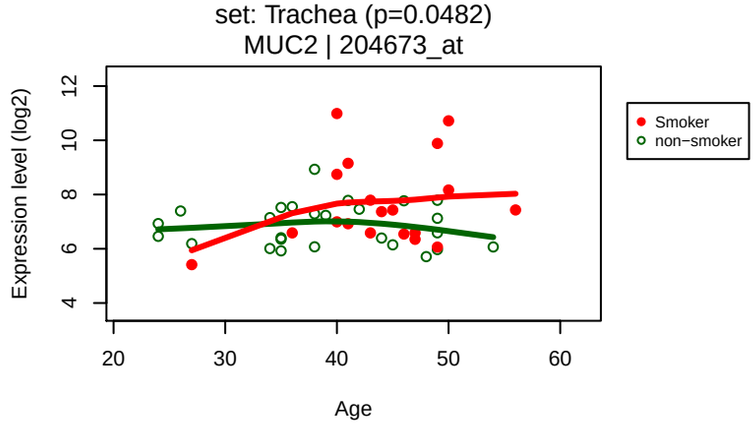

Gene expression ratio in Males
gene: MUC2 | 204673_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.114
p=0.0898
p=0.1688
p=0.1947
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC2 | 204673_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2431
p=0.0925
p=0.1887
p=0.4282
SmallAirway2
LargeAirway2
Trachea
Macrophage2

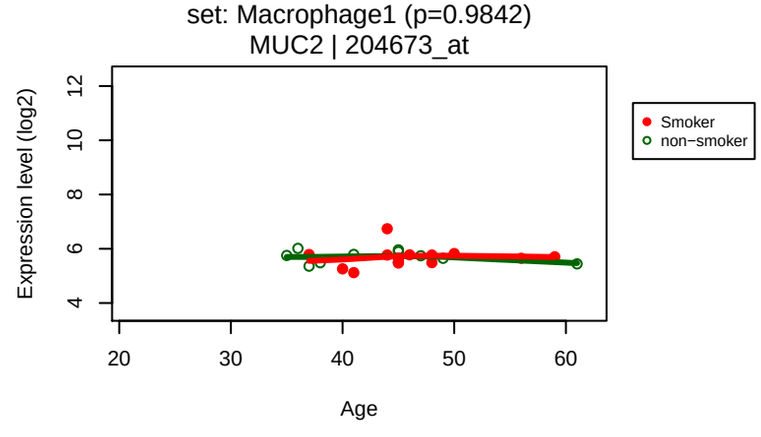

set: SmallAirway1 (p=0.3028)
MUC2 | 204673_at
Expression level (log2)
Age
Smoker
non−smoker

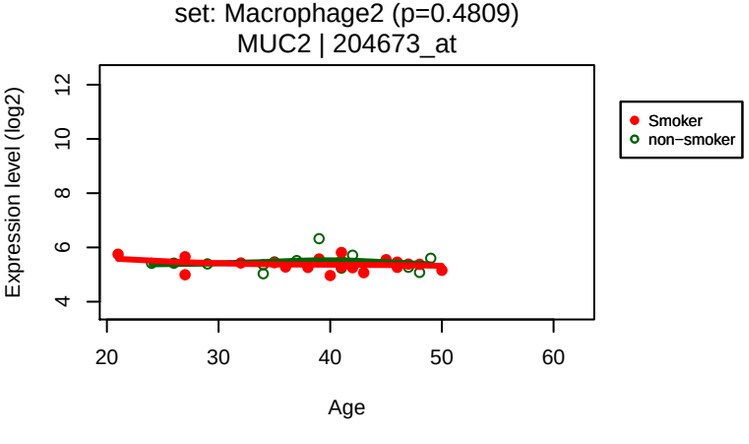

set: SmallAirway2 (p=0.0413)
MUC2 | 204673_at
Expression level (log2)
Age
Smoker
non−smoker

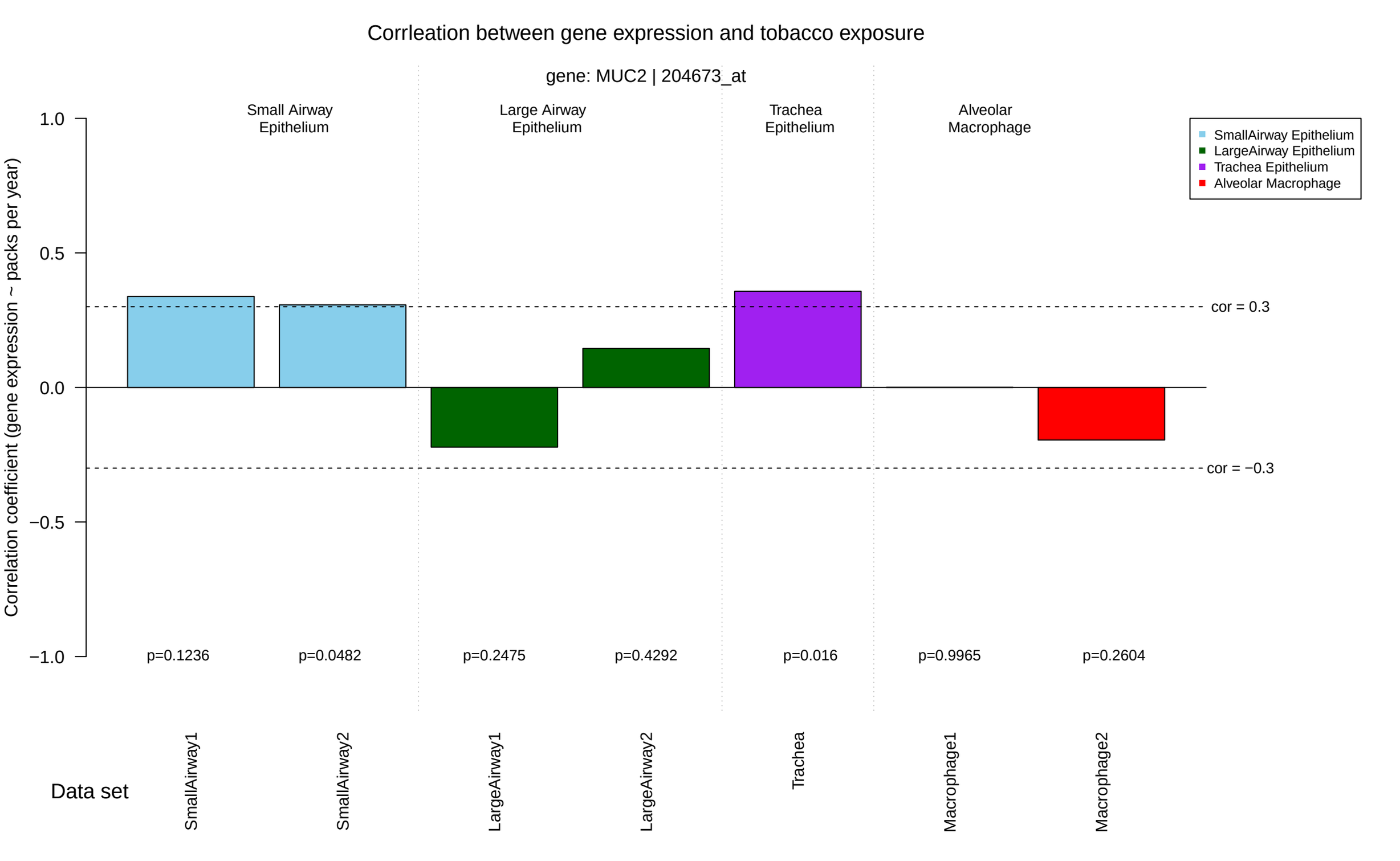

set: LargeAirway1 (p=0.7523)
MUC2 | 204673_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0282)
MUC2 | 204673_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0482)
MUC2 | 204673_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9842)
MUC2 | 204673_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4809)
MUC2 | 204673_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC2 | 204673_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1236
p=0.0482
p=0.2475
p=0.4292
p=0.016
p=0.9965
p=0.2604
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2